3-dars

Date. No.

Takrorlash: KLIB

KLIB - Python kutubxonasi bolib, data cleaning, visualization, va preprocessing uchun ishlatiladi

Asosiy imkoniyatlari:

- Missing value'larni topish
- Duplicate'larni o'chirish (Bir xil ma'lumotni takrorlanib qolishi)
- Column nomlarini tozalash
- Data type'larni optimallashtirish
- Correlation heatmap chiqarish
- Feature preprocessing qilish

- Correlation heatmap - datasetdagi ustunlar (features) bir-biri bilan qanchalik bog'liqligini rangli jadval orqali ko'rsatadigan grafik.

Machine Learning'da bu:

- qaysi feature'lar bir-biriga kuchli bog'langanini,
- qaysilari model uchun foydali yoki ortiqcha ekanini

aniqlash uchun ishlatiladi.

Masalan:

- study hours oshsa exam score ham oshsa → positive correlation
- age oshsa game time kamayib borsa → negative correlation

- Correlation qiymati

- qiymatlar ($-1 \leq r \leq 1$) -1 dan 1 gacha bo'ladi:

1 → juda kuchli positive bog'liqlik / -1 → juda kuchli negative bog'liqlik

0 → bog'liqlik yo'q

Date. No.

Heatmap nima?

Heatmap → rangli korinishdagi jadval

Masalan: Qizil → kuchli bog'liqlik / Ko'k → kuchsiz bog'liqlik

Python misol (seaborn bilan)

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv("data.csv")
corr = df.corr()
sns.heatmap(corr, annot=True)
plt.show()
```

klib bilan juda oson:

```
import klib
klib.corr_plot(df)
```

Bu automatik correlation heatmap chiqaradi.

#Nega kerak?

Machine Learning'da: ortiqcha feature'larni tapish, multicollinearity aniqlash, feature selection qilish, data'ni tushunish uchun juda foydali.

duplicate → bir xil ma'lumotning takrorlanib qolishi.

Nega yomon? → Duplicate data! modelni chalkashtiradi, accuracy'ni pasaytiradi, statistikani noto'g'ri qiladi, dataset hajmini keraksiz kattalashtiradi.

Date. No.

Pandas'da duplicate o'chirish:

y=x

```
import pandas as pd
df = pd.read_csv("data.csv")
# Duplicate qatorlarni o'chirish
df = df.drop_duplicates()
# Duplicate topish:
df.duplicated()
```

Data typelarni optimallashtirish: datasetdagi ustunlarning (int, float, string, bool va hokazo) turini to'g'ri va eng yengil formatga o'zgartirish degani.

Buni xotira (RAM) kam ishlatilishi, kod tezroq ishlashi, katta dataset bilan samarali ishlash uchun qilinadi.

KLIB → PYTHON da

```
pip install klib
import pandas as pd
import klib
df = pd.read_csv("data.csv")
# Data cleaning
clean_df = klib.data_cleaning(df)
# Missing values grafikasi
klib.missingval_plot(df)
```

Qachon ishlatiladi? → Dataset katta bolsa, Exploratory Data Analysis (EDA) qilishda, Tez preprocessing kerak bolsa.

Date. No.

2. DataCleaner / datacleaners nima?

Bu ham data preprocessing uchun Python package. U datasetni ML modelga tayyorlaydi.

Vazifalari:

- Missing values tozalash, • Duplicate row'larni olib tashlash,
- Normalization qilish, • Outlier'larni o'chirish,
- Clean dataframe qaytarish.

PYTHON da:

```
pip install datacleaners
from datacleaners import DataCleaner
import pandas as pd
df = pd.read_csv("data.csv")
cleaner = DataCleaner(df)
cleaner.handle_missing()
cleaner.remove_duplicates()
clean_df = cleaner.get_cleaned_data()
```

Farqi nimada?

klib → Visualization + preprocessing + EDA

DataCleaner → Oddiy va tez data cleaning

pandas → Manual control

scikit-learn → ML preprocessing pipeline

Date. No.

Built-in-function

- List
- Numpy
- Pandas

pipeline

List a=[1,2,3,4,5]

len(a) min(a) max(a) sum(a)
5 (nechta bor.) 1 (eng kichik) 5 (eng katta) 15 (yig'indi)

sorted (a) #tartiblab beradi ketma-ketlikdan kichik<katta

x=23.456
round(x) #butun qismini olib beradi
23
#Yaxlitlash u/n round(x,1) / round(x,2)
23.4 23.46

map + Lambda

list(map(lambda x: x*2))
list(map(lambda x: x*2, [1,2,3,4,5]))
Javob: [2,4,6,8,10]

filter + Lambda

list(filter(lambda x: x>2, [1,2,3,4,5])) #shart
Javob: [3,4,5]

not => tokari

list(filter(lambda x: x>0 and x%2==0, [1,2,3,4,5,-3,-6]))
[2,4]

Enumerate = enumerate (tartiblash)

for i, x in enumerate ('olma', 'beki', 'nok'): print(i,x)
0 olma
1 beki
2 nok

in / not in

Date. No.

Numpy

```
import numpy as np
import pandas as pd
        np.array([1,2,3])      # numpydan list yaratish
np.array([1,2,3]).mean()  # 2.0
np.array([1,2,3]).reshape(1,3)  # ([[1,2,3]])
np.array([1,2,3]).reshape(3,1)  # ([[1]
                                   [2]
                                   [3]])
np.random.randint(2)  # 2 gacha istiyoriy son
```

Pandas

```
df.info()
df.dropna(inplace=True)
df.fillna()n
#train/test/split
#LabelEncoder
```